Research
Interactive Poster Generator

By

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Introduction

Project Introduction

This project is a web-based tool designed to help graphic design students, beginners and other people who are just starting their journey in graphic design find inspiration and new ideas for their future projects.

The Interactive Poster Generator will combine user interaction with algorithmically generated visual compositions.

The project will offer two modes. The first mode will allow users to create their posters themselves by choosing colours, text, fonts and graphic elements and positioning them on the canvas in the way they want. This will allow users to experiment with different ideas and see how their compositions work without spending too much time putting all the elements together from scratch.

The second mode will be called Randomiser. With just one click, the website will use algorithms to generate a poster by randomly selecting colours, graphic elements, textures, fonts and other elements from its predefined library.

After creating their posters, users will be able to download them. Their creations will also be saved in a personal homepage gallery, where all previously created posters will be displayed in a grid layout.

This Research will explore 2 extensions as well to choose one and go on with it further. The Figma Poster Generator Plugin would allow people to generate posters within Figma and continue editing the selected composition. The Creative Assistant will offer design challenges and alternative visual directions to help users who find it difficult to start the project.

The main objective is to connect inspiration, experimentation and poster creation in one workflow.

Project Identity

- **Title:** Interactive Poster Generator
- **Type:** Interactive and generative design tool
- **Platforms:** Web browser, with one of proposed Figma plugin extension or Creative Assistant

- **Target audience:** Graphic design students, beginners and independent designers
- **Main features:** Manual poster creation, random generation, personal gallery and image export
- **Extensions:** Figma Poster Inspiration Generator or Creative Assistant

Core Design Pillars

Creative Freedom: Users should be able to modify generated compositions and make their own design decisions.

Controlled Randomness: The generator should create varied results while following basic rules for composition, readability and visual hierarchy.

Accessible Interaction: The interface should make the main functions understandable to users with limited design experience.

Creative Exploration: The system should encourage users to try different visual directions instead of immediately accepting the first result

Background Research

What is a poster?

A poster is a visual introduction to a product, event, brand or idea. They are used everywhere, and you can see them on streets, websites, Instagram posts and other social media platforms. Their main purpose is to grab the attention of anyone passing by or scrolling through content and hold that attention long enough to communicate the information displayed on it.

To achieve this, designers use the right composition, colour palette, typography and visual hierarchy. However, nowadays, it is not enough to simply put information together with a beautiful picture and call it a day. Modern poster design offers many different styles, extraordinary compositions, unexpected colour combinations and font pairings. All these elements work together to guide the viewer's eye across the canvas in the way the designer intended.

The same graphic design principles apply here as in publication layout, branding, or interface design; however, the poster compresses them into a format where nothing

can be hidden. There is no second page, no option to scroll, and no link to follow. Everything is visible at a glance.

What is an Art block

The art block or creative block is a common challenge for artists and designers at the same count. In simple words it’s when artists can't bring themself to create when before they have been able to generate ideas in their head and make them into something real. It feels like someone just stole that part of them and banned them from even looking at tools that they used every day for creating something special.

**Psychologists have proposed that the ability to combine existing ideas or information in new ways is more important for creativity than trying to produce ideas out of thin air. In support of this, it’s thought that our brains are constantly forming new connections between regions that process memories, knowledge, and challenges, without us even being aware of it. Every now and again, our brains bring together information in a meaningful way that results in a new idea, or a “eureka moment,” which appears to come out of nowhere.

And art block just blocks everything making people thing they cannot create anymore at the period it comes like that.

There could be many reasons why creating designs might be difficult: fear of starting the new project, fear of failing, fear of the success, fear of the judging from other people, fear of being not good enough, tight deadlines, not clear instructions about the project, not enough of the competence, keeping up with trends, etc.

Every design for everything starts with mood board. A mood board is a collage of images, patterns, videoframes, text and fonts that shows certain feeling for the project. Mood bords are widely used in different application of the art including branding, graphics, fashion, cinema, industrial and interior design. Sometimes, they are even used in nonvisual fields (e.g., music, fragrance design) to capture moods and feelings.

Generative Design

Generative design is a transformative approach to design that is reshaping our understanding of creation and innovation. It is a dynamic process that allows for the exploration of a vast range of design solutions, all of which adhere to defined constraints and are optimised for cost, material, and production methods.

The generative design is:

- Generate multiple design options
- Optimise each design option to meet specific design goals and parameters
- Significantly reduce the design repeating process
- Allow for the rapid exploration of multiple design alternatives
- Using generative design capabilities to improve the overall design process.

Traditional design methods need human input to reach required result. However with generative design these processes are automized and stabilised which make it a powerful tool.

Generative design offers a range of advantages that make it a powerful tool for professionals across many industries.

One of the key benefits of generative design is the optimization of the design process. By analysing input parameters—such as functional requirements, material properties, and manufacturing constraints—generative design tools can produce a multitude of valuable solutions and recommendations, thereby enhancing operational efficiency.

In addition to process optimization, generative design also facilitates:

- Reduced prototyping and manufacturing costs
- The creation of parts that require less material without compromising structural integrity
- Significant savings on materials, leading to lower overall production costs
- The creation of strong yet lightweight components (e.g., in the automotive industry), enabling an effective balance between conflicting industry requirements for strength and efficiency.

Popular generative design software includes Autodesk Generative Design, Creo Design Environment, and other computer-aided design (CAD) systems.

These generative design tools offer a range of features that make them indispensable in the development process. Key capabilities include:

- Parametric modelling;
- Topology optimization;
- Performance analysis;
- Automated design iteration.

These tools provide a comprehensive set of capabilities that optimise the design process and enhance operational efficiency.

Through a collaboration between Philippe Starck, Autodesk, and Kartell, generative design was employed to create innovative furniture; this project combined aesthetic concepts with manufacturing capabilities and demonstrated the potential of generative design to update traditional approaches to product design.

Randomness

Generally speaking, randomness is defined as a non-repeating, unbiased, non-patterned sequence of values.

"*In the visual world, randomness refers to elements that don't appear to follow any sort of intentional pattern or sequence.*"

A great example of randomness is when artist is throwing the paint into the canvas and let it spill as it wants. In graphic design though it wouldn't be as easy because every randomness should be created in its own order that wouldn't be too messy for the human to get a message it had to deliver.

Randomness serves a multitude of purposes in design. Firstly, it provides an element of surprise. When something falls outside of convention, it's far more likely to catch the viewer's attention.

Another benefit of randomness in design is that it can make it seem more human. While perfectly symmetrical and structured designs may look aesthetically pleasing, they can seem overly manufactured.



In this series by Paul Rand, a series of colourful dots create a stark contrast against a dark background. While the size of the dots remains the same in each individual frame, the colours and locations of them have been randomized. This is a great example of how having one consistent control element can help balance out the randomization.

Existing design tools

Poster Generator

https://keiver.dev/lab/poster-generator



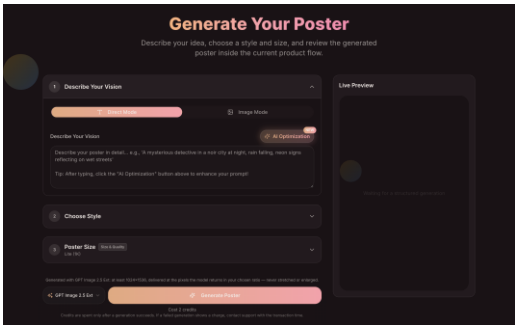
It generates abstract poster compositions using different visual styles, colour palettes and algorithmic patterns. It also supports custom text and downloading designs in formats such as SVG, PNG and JPG.

It demonstrates how a browser-based Randomiser could produce many variations from a predefined collection of styles and colours.

The difference is that user can't create the poster from scratch even though it still makes changes to those that website created.

AI Poster generation

https://poster.sh/



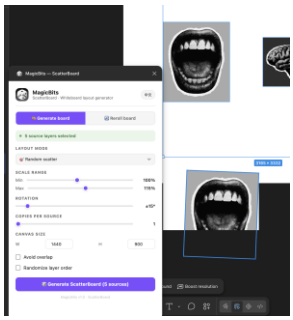
Poster.sh a website which makes prompt-based poster creation, followed by editing options for text, objects and layout by changing the prompt. User can also download the output of the poster which AI created.

It is a tool for creating a poster in a short time to visialise the idea of user.

I would'nt want to use AI for the project because it's important to me to make graphic designers think, create and improve their vision by using my product.

Figma Plugin for generating scatterboards

https://github.com/chenlicool/magicbits



This Plugin creates randomizes arrangements from selected layers, shapes and components. Its layout options include grids, clusters, scattered arrangements and other compositions.

Users could select a headline, images, shapes and textures, generate several poster layouts, and then continue editing the individual layers inside Figma.

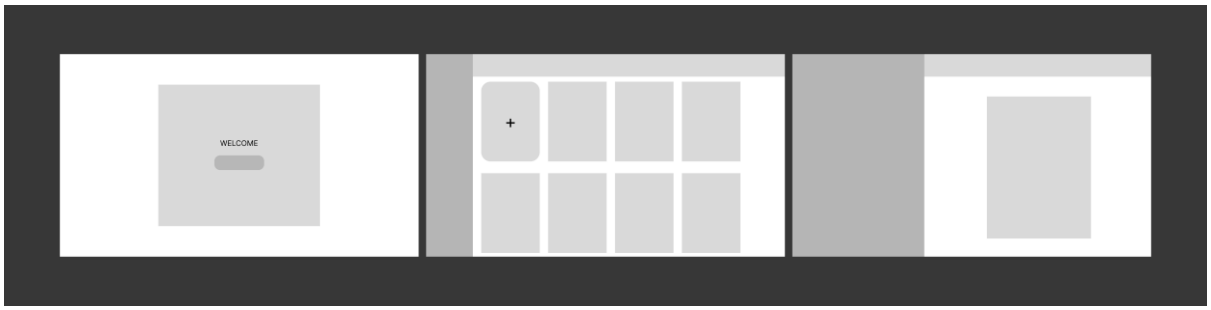
Core Functionality and Interaction Mechanics

Main User Journey

For the website:

Home page -> Creative mode -> Create the poster page -> Export/ Add the poster to their own library

Home page -> Randomiser mode -> Poster page -> Export/ Add the poster to their own library



For the Figma Plug in:

Run the Plug in -> Chose assets on a page-> Use options for changes in the plugin window-> Press “Create” button -> Result

For the Assistant:

Paste the assets into the window-> Chose options for changes-> Press “Create” button -> Result in the window

Creative Mode

Creative mode will allow the user to control every step of the process of creating simple version of poster that they had in their minds to check if their idea works or not on the canvas.

The user will be able to choose colours, assets, text, font, textures, change positions of graphic elements, etc.

After user had established every detail they wanted, the canvas should display them. Each element should have identified properties to be edited separately.

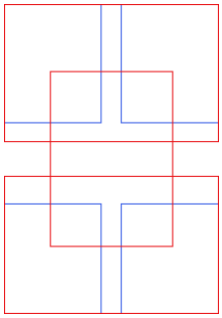
Randomiser Mode

Randomise mode will generate the composition by using everything from its library and positioning assets according to the randomised code.

The generation will work in those steps:

- Select graphic elements from the library;
- Select colors;
- Select font to the text that user will enter;
- Choose the size, rotation and position within the limits;
- Display the result on the canvas;
- Let user to edit details on the canvas;
- Give the ability to download the results;

In this mode user will receive a challenge and will be able to edit the result to their liking to create the masterpiece in another software (or Figma if it is a plugin).



(Position options for random mode)

Figma Poster Generator (plugin)

The Figma poster generator will be a plugin that will help users to find inspiration using their own graphic elements to combine them easier into the poster idea and keep working in Figma using the result as a reference.

It should be a compact little widow which will give the user options to apply the colour, fonts, sizes and rotation on the window to apply it to the frame on the Figma workspace. In a randomise mode user can chose assets on a frame and a plugin will choose the position, rotation, colours and font by itself scattering it across the frame to give a start point for the user.

Creative Assistant

Creative assistant will look like the little window hovering over while user will browse anything they need. It will allow user to find the idea of what they are looking for to create something of their own.

Assistant will combine the research of the needed assets for the project and unique reference they will be able to create with Creative Assistant. It will reduce the time and extra effort by cutting the need to switch between the websites to refresh the memory of the created reference.

Interface, User Experience and Navigation

Welcome Screen

Welcome page will introduce the user to the name of the project and give the clear instruction for the entry. It should contain the title, short instruction and the entry button.

This minimal style page will not overwhelm the user and will give the easy start to start creating their own inspiration.

Personal Gallery

Personal gallery page will be a page with a grit of previous poster user had made and an empty canvas on the beginning of it which would allow user to start the work of a new project.

The grid of previous posters will allow to look through it and export it multiple times on a devise user uses.

Poster Workspace

This page will have the canvas to review changes and on the right side there will be buttons, bars and different other things with which help user will be able to create the poster.

In the centre will be a rectangle of the white space where will be shown every detail user adds to get their poster done and where user will be able to position graphic elements from the library.

Design controls will include: square with colours which user will be able to pick; font choice and the size of the text; options of different graphic elements and textures for the design.

The layout will depend on the order of a choice of graphic elements of user.

Plugin / Assistant Interface

Plugin and assistant will use a compact interface. They will show the same layout of controls as on a website.

The only difference between plugin and assistant will be that in the Figma there won't be any need for the canvas window because in that tool user will be able to create the frame by themself according to their need.

In assistant window whatsoever will be a tiny window under all controls but without any possibility to change any position of graphic elements because of the need to maintain the size of a small window to not distract user from different tabs. Instead, there will be either random position already built in, or specific location attached to every asset individuate.

User Experience Requirements

The requirements for the good User Experience:

- Controls must have understandable labels to not make user confused and not make him spend more time to figure out what is responsible for what
- Website should not concentrate only on colours. The interface should have the contrast
- Important information should be visible

- Canvas should show every change across the project process
- User should be able to leave the website/plugin/assistant without losing the progress
- All actions available should remain consistence across website/plugin/assistant

Technical Scope and Requirements

Website Technologies

- The website will need to use HTML, CSS and JavaScript.
- The Figma plugin: Figma desktop app, Visual Studio Code (install TypeScript and Node.js).
- The Assistant (Extension): HTML, CSS and JavaScript.

HTML will provide the base/structure, CSS will shape it to something useable and pleasant for the eye, JavaScript will make the website/plugin/assistant functional, generative and interactive aspects.

Data and Generation Rules

Every poster projects the website/plugin/assistant hast to collect and save data for colours, font, sizes, positions and graphic elements.

The Randomise will use prepared libraries and position setting beforehand to create variations. Each graphic element should be quiet identified to be able for the user to change positions of the elements after the code will place them on canvas.

Saving and Exporting

The website and assistant supposed to have saving and exporting the result in PNG or JPG format and save poster into the website gallery so user could save it on their page of the website to have a possibility to have a look on their creations whenever they need.

Figma though will not need this function because this tool already was made to save whatever project they created.

Technical Constraints

The biggest challenge is to keep the readability on a small space of the plugin and assistant windows, keep all the elements visible on the canvas as well. On the website it will be the challenge to give the possibility to save every created poster on the drive of the website.

The project has to privide reliable functions and user experience over having bigger amount of features. The performants requirements and technical limits will be updated and fixed after the base of the project will run smoothly.

Prototype Scope and Development Priorities

Project will be in created in stages:

1. **Project Design:** Create simple wireframes for the better understanding what exactly will be made, add colours and fonts to the design.

2. **Core of a project:** Will need to code the base of a project and basic function like “create” and “randomise” buttons, put in the colour pallet, connect google fonts resorts to the project, make it possible to change sizes of the text.

   For the website/assistant: Create the asset library to make it possible to choose the graphic elements, create the canvas to show the result, make the function to move the elements around the canvas.

3. **User Interface & Randomise function:** Make better interface, update the functions to make them work more precise, make the randomise function work.

4. **Saving and export for the website/assistant:**
   Create an ability to save the posters into the user’s library and export it as PNG/JPG files.

Testing and Evaluation

The project will need to be tested at least by 3 different people to get to know what works and what needs upgrades. It will give valuable feedback with which it will give the understanding about user experiences, struggles users can face and will give a hint in which direction the project will need to be upgraded.

- The testing will be completed by at least 3 users.

- Users will be asked followed up questions.
- Answers will be noted and grated afterwards.
- Users will be asked to give some suggestions what will upgreate in their opinion the project.

Conclusion

After doing the research, I lean more to the Figma Plugin version. Tools like Keiver's Poster Generator and MagicBits Scatter Board demonstrate how randomise can help beginner users to find their own inspiration according to their choices. However, I would like to combine these abilities with a possibility to use own different assets without any limitations about their option and continue editing the result in the familiar workspace.

The main reason for a Figma plugin to be as a base for this project is a possibility to work with existent personal assets without the need to move between applications or a struggle to choose only something similar to what they want to see in their own projects. User could select every little aspect in there and experiment more with varieties of the possible posters and inspirations for some other projects. It would give more freedom for user to play around for their projects without switching to other tools of software.

Even though Generative Assistant would be able pretty much the same, but it would put some restrains for the creativity process as like asset library. It would be hard to make it work with the number of functions it would need to hold in a small window including with interactive canvas. It would have to hold smaller buttons and text which could create some challenges for the user to use it and abandon the idea of this type of project in the end.

Including all challenges that came up in the stage of the research I lean more to the Figma plugin. And the goal of this project is to create the tool which will help to get through challenges in the start points of the design points, let users to experiment with different compositions and turn the idea into something more real.

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